MSCF Probability: Homework #2 Solutions

Each question is worth four points.

1. Let X be the number of upward moves in 7 periods.
   $\Rightarrow\ X \sim \text{Bin}(7, 0.4)$.
   In order to achieve $t_7 \geq 32$, there must be, among the first seven moves, at least three more upward moves than downward moves. Thus, X must be either 5, 6, or 7 (corresponding to 2, 1, or 0 downward moves, respectively).

$$P(S_7 \geq 32) = P(X \geq 5) = \sum_{x=5}^{7} \binom{7}{x} (0.4)^x (0.6)^{7-x} = 0.096256$$

2. Let B be the event that the drug is beneficial and let X be the number of colds a person has in a year.
   We are given:

$$\begin{aligned} P(B) &= 0.6 \\ P(X = 2|B) &= \frac{e^{-3}3^2}{2!} = 4.5e^{-3} \\ P(X = 2|B^c) &= \frac{e^{-5}5^2}{2!} = 12.5e^{-5} \end{aligned}$$

   We need to find $P(B|X = 2)$. Using Bayes,

$$P(B|X = 2) = \frac{P(X = 2|B)P(B)}{P(X = 2|B)P(B) + P(X = 2|B^c)P(B^c)} = \frac{4.5e^{-3}(0.6)}{4.5e^{-3}(0.6) + 12.5e^{-5}(0.4)} \approx 0.8$$

3. Since

$$| x |= \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

   $f_X(x)$ can be written in the following way:

$$f_X(x) = \begin{cases} .5e^{x} & x < 0 \\ .5e^{-x} & x \geq 0 \end{cases}$$

   Now

$$\begin{aligned} P(2.5 \leq | X | \leq 3.5) =& P(-3.5 \leq X \leq -2.5 \text{ or } 2.5 \leq X \leq 3.5) \\ =& P(-3.5 \leq X \leq -2.5) + P(2.5 \leq X \leq 3.5) \\ =& \int_{-3.5}^{-2.5} .5e^{x} dx + \int_{2.5}^{3.5} .5e^{-x} dx \\ =& [.5e^{x}]_{-3.5}^{-2.5} + [-.5e^{-x}]_{2.5}^{3.5} \\ =& .5(e^{-2.5} - e^{-3.5}) - .5(e^{-3.5} - e^{-2.5}) \\ =& e^{-2.5} - e^{-3.5} \\ \approx& .052 \end{aligned}$$

4. Let X be the lifetime of the component (in hundreds of hours).

$$\Rightarrow X \sim exp(\beta = 1/10).$$

$$P(X \leq 11|x > 9) = 1 - P(X > 11|x > 9) = \text{(memoryless)}$$

$$1 - P(X > 2) = F_X(2) = 1 - e^{-2/10} \approx 0.1813$$

If $X \sim U(8, 12)$, the memoryless property is not satisfied, and therefore,

$$P(X \leq 11|X > 9) = \frac{P(9 < X \leq 11)}{P(X > 9)} = \text{(uniform probabilities)} = \frac{2/4}{3/4} = .667$$

5. Let $T = \min(X, Y)$. Then, $P(T \leq 0) = 0$, and for $t > 0$,

$$\begin{aligned} P(T \leq t) &= 1 - P(T > t) \\ &= 1 - P(X > t \text{ and } Y > t) \\ &= 1 - P(X > t)P(Y > t) \text{ since } X, Y \text{ independent} \\ &= 1 - e^{-\lambda_1 t} e^{-\lambda_2 t} \\ &= 1 - e^{-(\lambda_1 + \lambda_2)t} \end{aligned}$$

Hence, $T \sim \text{Exp}(\lambda_1 + \lambda_2)$.